Digital Communication Systems and Sampling

1.1 Digital transmission and receiver decisions

Key point. Use f in hertz and $\omega = 2\pi f$ in radians per second. Sampling fixes the time instants, whereas quantization restricts amplitude levels. These operations must not be interchanged.

Technical reference: MIT OpenCourseWare, 6.450 lecture notes.

Definition / Digital communication

Digital communication sends information using a finite set of symbols. In a binary system, the information is represented by bits: 0 and 1.

A voice message begins as sound. A microphone changes the sound into a voltage. The system represents this voltage by numbers, sends those numbers, and uses them to recover the message. The receiver does not need an exact copy of every transmitted voltage. It must identify the correct symbols, then interpret their agreed meaning.

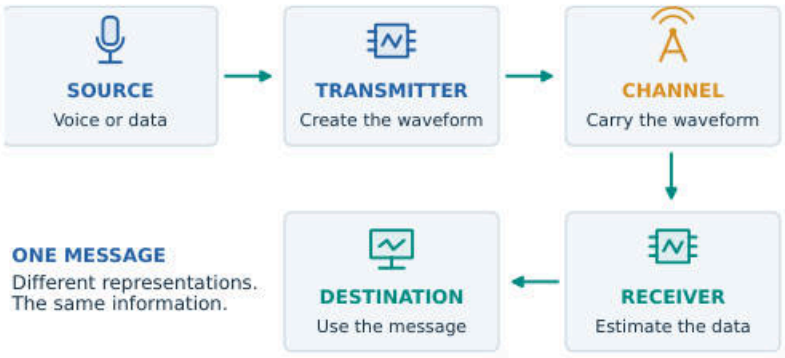


Figure 1.1: Follow the arrows from the source to the destination.

Source. Produces information, such as voice, text or a photograph.

Transmitter. Changes the information into a waveform suitable for the channel.

Channel. Carries the waveform through a cable, optical fiber or free space.

Receiver. Uses the received waveform to estimate the transmitted information.

Destination. Uses the recovered information, for example through a loudspeaker or a screen.

Key point. Bits are labels. The channel carries a physical waveform, such as a voltage, a light signal or a radio wave.

Check the concept

In a voice call, what is the source? **Answer:** The speaker's message. The microphone converts the sound into an electrical signal.

The basic path is now clear. A practical system can include extra blocks to make transmission more efficient or more reliable. In the diagram, coding prepares the data before transmission. Decoding reverses the representation after the receiver has made its decisions.

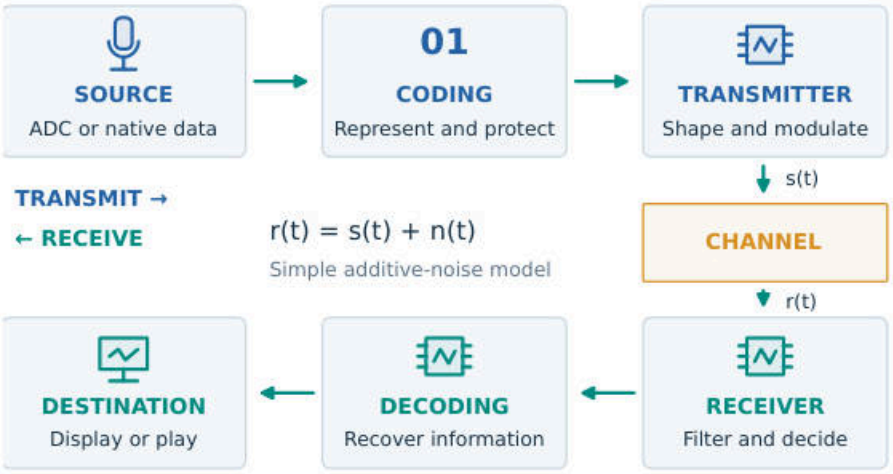


Figure 1.2: A more detailed link. The upper row prepares and transmits information. The lower row receives, decodes and delivers it. ADC means analog-to-digital converter.

- **Source coding:** represents information efficiently, for example by compressing an image.
- **Channel coding:** adds structured extra bits to help detect or correct errors.
- **Pulse shaping:** creates a controlled waveform from the symbols.
- **Carrier modulation:** places information on a carrier when a passband channel is used.
- **Detection:** decides which symbol was most likely sent.

> **Key point.** Source coding and channel coding have different purposes. Compression reduces predictable redundancy. Error-control coding adds useful redundancy. A baseband wired link does not necessarily need radio-frequency (RF) modulation.

Simple example

A receiver observes $+0.8$ V. The system uses $+1$ V for bit 1 and -1 V for bit 0. With a zero threshold, the receiver decides **1**, because $+0.8$ V is positive.

1.2 Sinusoidal signals, frequency and time units

Sinusoidal signals: meaning of the symbols

A sinusoid describes how a voltage changes with time. First identify its amplitude, frequency and phase. Only then calculate its period or its value at a particular time.

Standard sinusoid

$$x(t) = A\cos(2\pi ft + \phi) = A\cos(\omega t + \phi)$$

$x(t)$ is voltage in volts (V). A is peak amplitude in V. t is time in s. f is frequency in hertz (Hz), or cycles/s. ϕ is phase in radians. π is the constant 3.14159...; ω is angular frequency in rad/s.

The cosine argument is an angle in radians. For a sine, replace cos by sin. Amplitude changes vertical size; frequency changes the number of cycles per second. Phase determines the position of a cycle relative to t = 0.

Angular frequency and period

$$\omega = 2\pi f; f = \omega/(2\pi); T = \frac{1}{f}$$

T is the cycle period in s. One complete cycle contains 2π radians.

Convert angular frequency to Hz before applying a sampling rule. If $\omega = 400\pi$ rad/s, then f = 200 Hz and T = 0.005 s. Do not confuse cycle period T with sampling interval T_s.

Evaluating one signal value

$$\theta = 2\pi ft + \phi; x(t) = A\cos(\theta)$$

θ is the instantaneous angle in radians.

Convert the requested time to seconds, calculate θ, and evaluate the trigonometric function in radians. cos(0) = 1, cos(π/2) = 0 and cos(π) = -1. A zero crossing is a valid voltage value.

Example 1.1

For x(t) = 2cos(2π × 500t) V, find amplitude, frequency and period.

Find: Amplitude, frequency and period.

SOLUTION

Read A and f by comparison; calculate T = 1/f.

$$x(t) = A\cos(2\pi ft)$$

$$A = 2\,\text{V}.$$

$$\omega = 2\pi f$$

$$f = 500\,\text{Hz}.$$

$$T = \frac{1}{f}$$

$$T = 1/500 = 0.002\,\text{s} = 2\,\text{ms}.$$

Answer. Amplitude 2 V; frequency 500 Hz; period 2 ms.
Amplitude measures vertical size. Period measures the time for one cycle.

Example 1.2

A waveform repeats every 4 ms. Find its frequency.

Find: Frequency in hertz.

SOLUTION

Frequency f = 1/T, with T in seconds.

$$T = 4 \times 10^{-3}\,\text{s}$$

$$4\,\text{ms} = 4/1000 = 0.004\,\text{s}.$$

$$f = 1/T$$

$$f = 1/0.004 = 250\,\text{Hz}.$$

$$fT = 1$$

Check: 250 cycles fit into one second.

Answer. 250 Hz.

Do not substitute 4 directly as seconds. That would produce an incorrect frequency of 0.25 Hz.

Example 1.3

Find x(0) and x(0.25 ms) for x(t) = cos(2π × 1000t) V.

Find: Signal values at the two specified times.

SOLUTION

Substitute the requested time in seconds into the cosine.

$$x(0) = A\cos(0)$$

$$x(0) = 1\,\text{V}.$$

$$t_s = t_{\text{ms}}/1000$$

$$0.25\,\text{ms} = 0.00025\,\text{s}.$$

$$\theta = 2\pi f t$$

$$\theta = 2\pi \times 1000 \times 0.00025 = \pi/2.$$

$$x(t) = A\cos(\theta)$$

$$\cos(\pi/2) = 0.$$

Answer. $x(0) = 1\,\text{V}; x(0.25\,\text{ms}) = 0\,\text{V}$.

A zero crossing is a normal signal value, not the absence of sampling.

1.3 Uniform sampling and sample indices

Definition / Sampling

Sampling measures a continuous-time signal at selected instants. Uniform sampling uses the same time interval between consecutive measurements.

$$T_s = \frac{1}{f_s}, \qquad x[n] = x(nT_s).$$

- f_s: sampling rate in samples per second.
- T_s: time between samples, in seconds.
- n: integer sample number, $0, 1, 2, \ldots$
- $x[n]$: value of the sample taken at time nT_s.

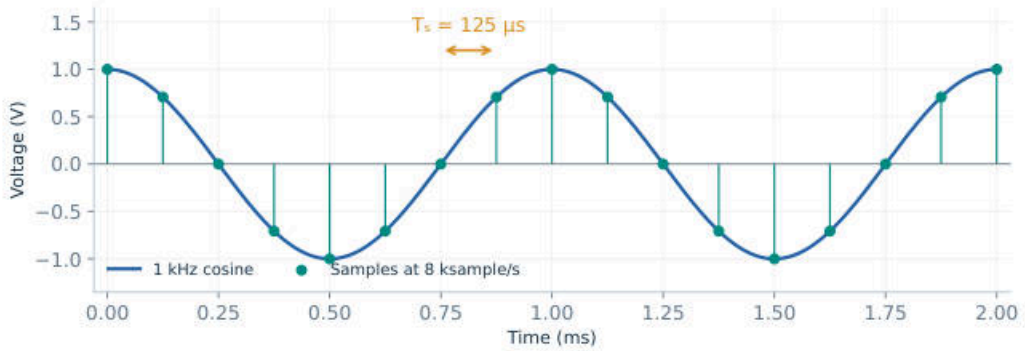


Figure 1.3: The continuous curve is measured at the marked instants.

Example 1.4

A signal is sampled at 8000 samples/s. Find the interval between samples.

SOLUTION

Take the reciprocal of the rate, then convert seconds to microseconds.

$$T_s = \frac{1}{f_s} = \frac{1}{8000}\,\mathrm{s} = 125\,\mu\mathrm{s}.$$

Answer. One sample every 125 μs. Sampling alone does not quantize its amplitude.

Sample indices and a finite record

The integer index n selects the measurement time $t_n = nT_s$; write $x[n] = x(nT_s)$. For an aligned record of duration D starting at zero, $[0, D)$ contains D/T_s samples and $[0, D]$ contains one additional endpoint sample. These counts assume that D/T_s is an integer.

Example 1.5

Count samples at $f_s = 1000\,\mathrm{sample/s}$ over $0 \le t < 10\,\mathrm{ms}$, then include the final endpoint.

SOLUTION

The sampling interval and half-open count are

$$T_s = 1/f_s = 1\,\mathrm{ms}, \qquad N = D/T_s = 10.$$

The instants are $0, 1, \ldots, 9\,\mathrm{ms}$. Including 10 ms adds one sample.

Answer. 10 samples for $[0, 10\,\mathrm{ms})$, or 11 for $[0, 10\,\mathrm{ms}]$.

Watch sampling happen

Use $x(t) = \cos(2\pi\,1000t)$ V with t in seconds. Choose $f_s = 4000$ sample/s, so $T_s = 0.25$ ms.

$$x[n] = x(nT_s) = \cos(\pi n/2)\ \text{V}.$$

Read the curve. The top panel is continuous in time. The clock chooses instants, not amplitude levels. The first five sample values are $[1, 0, -1, 0, 1]$ V.

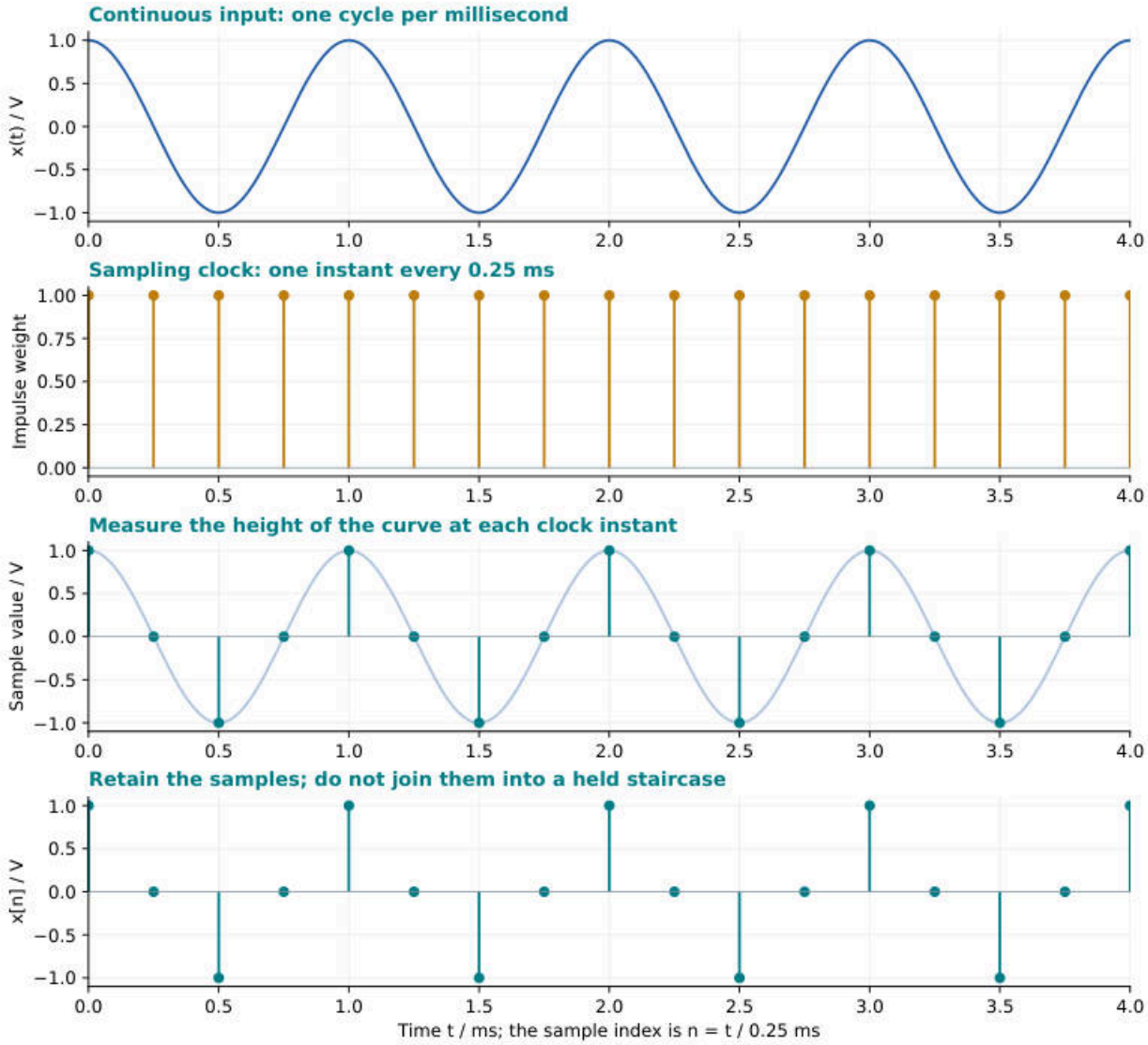


Figure 1.4: Follow one time position through the four panels. The measured height becomes a sample value.

Keep the distinction. Sampling discretizes time, not amplitude. Quantization is separate. Stems indicate sample values, not finite-width physical pulses. In the ideal model $x_s(t) = \sum_n x[n]\delta(t - nT_s)$, $x[n]$ denotes impulse weight. A practical sampler has a finite aperture.

1.4 Sampling, quantization, encoding and data rate

Definition / Analog signal

An analog signal represents information using a continuously varying quantity, such as microphone voltage.

Before an analog signal can be represented by bits, it passes through three different operations.

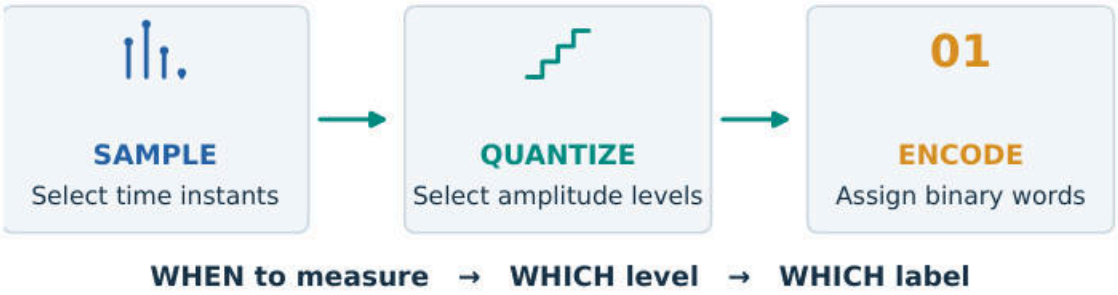


Figure 1.5: Sampling chooses times, quantization chooses levels, and encoding assigns labels.

Sample. Measure the voltage at regularly spaced times.

Quantize. Replace each measurement by the nearest allowed level.

Encode. Write the selected level as a binary word.

Together, these operations form an **analog-to-digital converter (ADC)**. Sampling preserves the measured value at a chosen time. Quantization introduces rounding. Encoding changes the label, not the chosen level.

Example 1.6

A sample has value 0.62 V. The allowed levels are 0, 0.25, 0.50 and 0.75 V, with labels 00, 01, 10 and 11. What word is produced?

SOLUTION

Measured sample: 0.62 V. Four allowed levels.

Find the nearest level. The distances to 0.50 and 0.75 V are 0.12 and 0.13 V. Choose 0.50 V.

Read its label. The label of 0.50 V is 10.

Answer. The transmitted label for this sample is **10**.

Definition / Quantization

Quantization replaces each sample value by one of a finite number of allowed amplitude levels.

Definition / Encoding

Encoding assigns a bit word to the selected level. With b bits per word, there are $L = 2^b$ possible words.

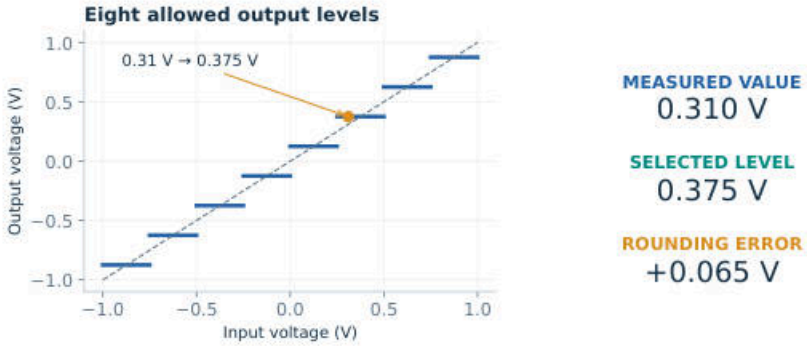


Figure 1.6: Quantization rounds sample values. The difference from the original value is the quantization error.

For a uniform quantizer with range $V_{\min}$ to $V_{\max}$ and L equal bins,

$$\Delta = \frac{V_{\max} - V_{\min}}{L}.$$

Here Δ is the bin width. For midpoint rounding without overload, the error magnitude is at most $\Delta/2$.

Example 1.7

A uniform 3-bit quantizer covers -1 to $+1$ V. Find the number of levels, bin width and maximum rounding error.

SOLUTION

$b = 3$, $V_{\min} = -1$ V, $V_{\max} = 1$ V.

Count the levels. $L = 2^3 = 8$.

Divide the voltage range. $\Delta = [1 - (-1)]/8 = 0.25$ V.

Find half a bin. $|e_q| \leq 0.25/2 = 0.125$ V.

Answer. 8 levels, bin width 0.25 V, and error magnitude at most 0.125 V for samples within the quantizer range.

Binary words and transmitted bit rate

Encoding labels a selected quantizer level with a bit word. The same level-to-word mapping must be used at the transmitter and receiver.

Minimum word length

$$2^b \geq L; b_{\min} = \lceil \log_2 L \rceil$$

L is the number of levels; b and $b_{\min}$ are integer bits per sample. $\log_2$ is the base-two logarithm. ceil means round upward to the next integer.

Choose the smallest b that provides enough distinct words. For 10 levels, 3 bits give only 8 words; 4 bits give 16 words. Some words may remain unused.

Natural-binary index

$$i = b_2 2^2 + b_1 2^1 + b_0 2^0$$

i is the level index. Each digit b_2, b_1 or b_0 is 0 or 1. Their place weights are 4, 2 and 1. These digits differ from b, the word length.

In a 3-bit word, 101 represents index 4 + 0 + 1 = 5. The index identifies the level under the agreed mapping; it is not itself a voltage.

Payload and overhead

$$R_{\text{payload}} = f_s b; R_{\text{added}} = \alpha R_{\text{payload}}; R_{\text{total}} = (1+\alpha) R_{\text{payload}}$$

Rates R are in bit/s; f_s is in sample/s; b is in bit/sample. α is added overhead as a fraction of payload. Multiply sample rate by bits per sample. If added overhead is 25% of payload, $\alpha = 0.25$. This definition differs from overhead specified as a fraction of total transmitted bits.

Bit interval

$$T_b = \frac{1}{R_{\text{total}}}; N = Mb$$

T_b is the time for one bit in s. M is the sample count and N is their total payload bit count.

T_b differs from sample interval T_s: one sample can require several bit intervals. NRZ means non-return-to-zero; a line code converts the bit words into a physical waveform.

Example 1.8

The following problem concerns assigning binary words to quantizer levels or transmitting the resulting sample data. An ADC samples at 8000 sample/s and uses 8 bits per sample. Find its uncoded payload rate.

Find: Uncoded payload bit rate.

SOLUTION

$$R = f_s \times b$$

$$R = 8000 \times 8 = 64000\,\text{bit}/s.$$

$$1\,\text{kbit/s} = 10^3\,\text{bit/s}$$

$$64000/1000 = 64\,\text{kbit/s}.$$

$$[R] = \frac{\text{samples}}{\text{s}} \frac{\text{bits}}{\text{sample}} = \text{bit/s}$$

Answer. 64 kbit/s.

This excludes framing, channel coding, packet headers and other overhead.

1.5 Communication media, propagation and signal impairments

Definition / Channel

A communication channel is the path through which a transmitted signal reaches a receiver.



Figure 1.7: Guided channels confine the signal to a physical path. Unguided channels transmit waves through space.

Channel	What carries the signal?	Familiar example
Twisted pair	Electrical signal on a pair of wires	Copper network cable
Coaxial cable	Electrical signal between conductor and shield	Antenna feeder
Optical fiber	Light in a glass fiber	Optical data link
Terrestrial microwave	Radio wave between ground antennas	Point-to-point link
Satellite	Radio wave via a satellite	Satellite television
Broadcast radio	Radio wave to many receivers	Radio broadcasting

Key point. Choose a channel by its required distance, data rate, interference environment and installation conditions. There is no single best medium for every link.

Check the concept

Is optical fiber a wireless channel because it carries light? **Answer:** No. The fiber guides the light, so it is a guided channel.

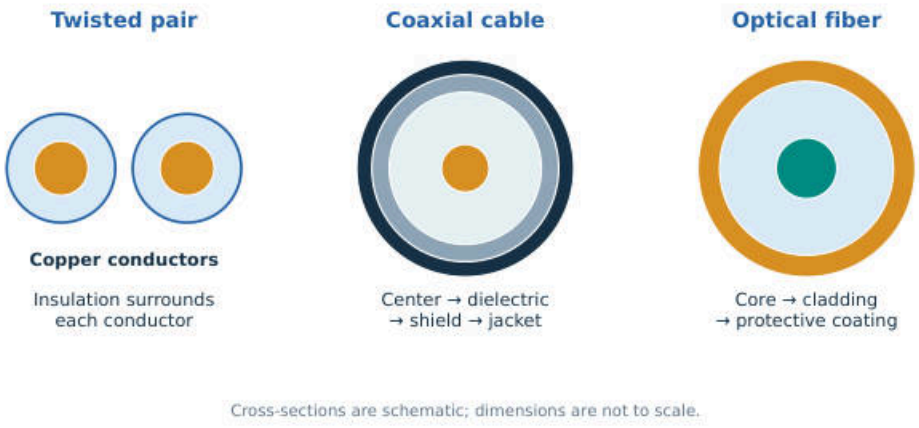


Figure 1.8: The cross-sections show the layer order in three common media. Dimensions are schematic, not to scale.

Twisted pair. Two insulated conductors are twisted together. In a balanced link, the receiver measures the voltage difference between them. Twisting helps reduce interference pickup.

Coaxial cable. A central conductor is surrounded by insulation and an outer conducting shield. The geometry helps confine the electromagnetic field.

Optical fiber. A core and cladding guide light. The optical signal is not affected by electromagnetic pickup in the way a copper signal is.



Figure 1.9: A real LC optical-fiber connector (photo: Adamantios, CC BY-SA 3.0). Its ferrule holds the fiber accurately so that two fiber ends can be aligned.

Compare before choosing

Copper connections carry electrical signals and need suitable protection from interference. Fiber is useful where electrical isolation or long-distance optical transmission is needed. Fiber connectors must be clean and correctly aligned.

> **Key point.** Fiber still has losses and signal distortion. Immunity to electrical interference does not mean that every optical link is error-free.

- **Terrestrial microwave:** directional antennas connect two ground sites.
- **Satellite link:** a satellite relays a signal between ground stations.
- **Broadcast radio:** one transmitter serves many receivers.



Figure 1.10: Dish antennas at the Goldstone complex (photo: NASA/JPL-Caltech, PIA23214). A reflector directs radio energy toward the feed when receiving and forms a narrow beam when transmitting.

Any channel can change the transmitted waveform. Three effects are especially important.

Definition / Attenuation

A reduction in signal amplitude or power. The received signal becomes weaker.

Definition / Distortion

A change in waveform shape, for example because different frequency components experience different gains or delays.

Definition / Noise

An unwanted random contribution added to the received signal.

Key point. A weak signal, a distorted signal and a noisy signal are different problems. Compare their shapes in the attenuation, distortion and noise figure in the following section before using any equations.

Transmission duration and propagation delay

Sending N bits at rate R_b takes $t_{\text{tx}} = N/R_b$. Propagation over distance d at speed v takes $t_{\text{prop}} = d/v$. Ignoring processing and queueing, the last bit arrives at $t_{\text{tx}} + t_{\text{prop}}$ after the first bit starts transmission. Increasing bit rate shortens serialization, not propagation travel time.

Example 1.9

A 100-bit block travels at 1 Mbit/s through a link with propagation delay 10 μs. Ignore processing and queueing.

SOLUTION

First calculate the time needed to send all bits.

$$t_{\text{tx}} = \frac{N}{R_b} = \frac{100}{10^6} = 100\,\mu\text{s}.$$

Then add the propagation time for the last bit.

$$t_{\text{last}} = t_{\text{tx}} + t_{\text{prop}} = 100 + 10 = 110\,\mu\text{s}.$$

Answer. The first bit starts arriving at 10 μs and is completely received at 11 μs. The complete block is received at 110 μs.

1.6 Attenuation, distortion and noisy decisions

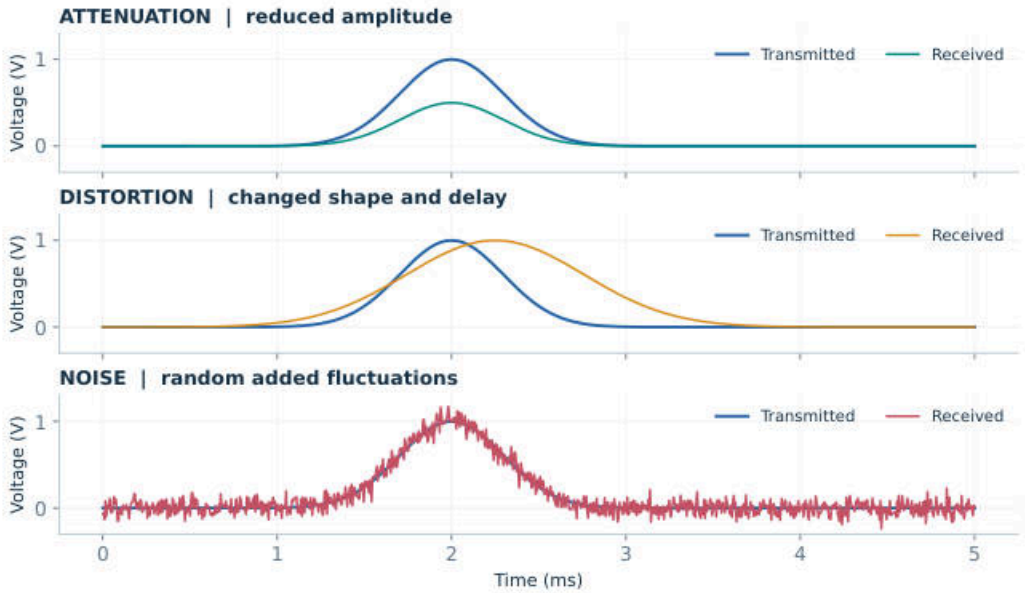


Figure 1.11: Compare each received waveform with the reference: smaller amplitude, changed shape, or random fluctuations.

A simple noise-only model is

$$r(t) = s(t) + n(t).$$

Here $s(t)$ is the transmitted signal, $n(t)$ is additive noise, and $r(t)$ is the received signal. A simple attenuation-and-noise model is $r(t) = as(t) + n(t)$, where $0 < a < 1$ is an amplitude scale factor.

Example 1.10

A transmitted sample is $+1$ V. The channel scales it by 0.5 and adds -0.1 V of noise. Find the received sample.

SOLUTION

$s = 1$ V, $a = 0.5$, $n = -0.1$ V.

Apply attenuation. $as = 0.5 \times 1 = 0.5$ V.

Add the noise. $r = 0.5 + (-0.1) = 0.4$ V.

Answer. The receiver observes 0.4 V. A zero-threshold binary detector would still decide 1.

1.7 Nyquist rate: single tones, sums and products

Definition / Baseband bandwidth

For this lecture, a real baseband signal has bandwidth B when its spectrum has no content outside $-B$ to $+B$. Thus B is its highest positive frequency.

To preserve a band-limited baseband signal, use

$$\boxed{f_s > 2B}.$$

The boundary $2B$ is called the **Nyquist rate**. Its reciprocal $1/(2B)$ is the **Nyquist interval**. Practical sampling uses a rate above the boundary to allow a real filter transition band.

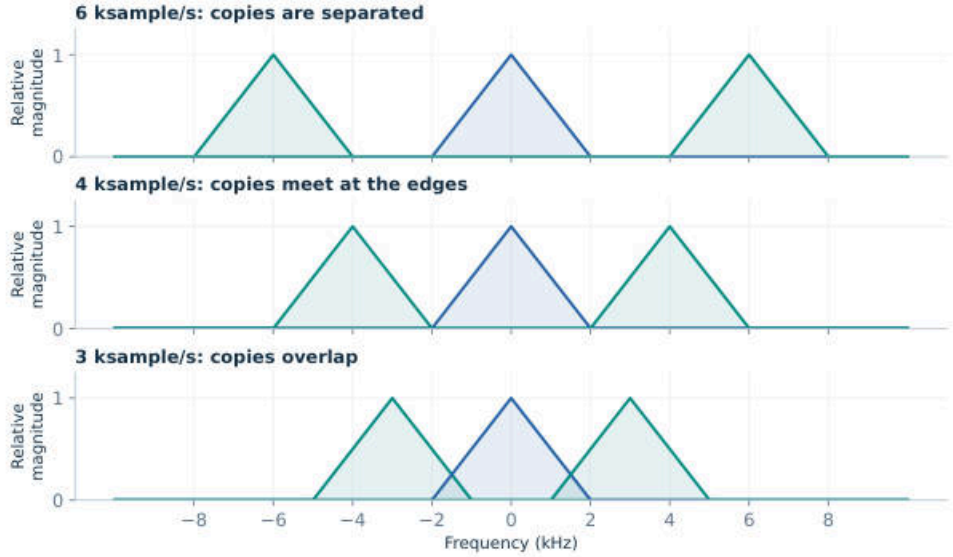


Figure 1.12: For a 2 kHz baseband limit, the repeated spectra are separate, touching, or overlapping. Only the central copy and its two neighbors are shown. Heights are normalized.

Key point. The equality $f_s = 2B$ is a limiting case, not a safe choice for every signal. A sine wave exactly at the boundary can produce all-zero samples for a particular phase.

Three cases

$f_s > 2B$: separated copies. $f_s = 2B$: copies meet at the boundary.

$f_s < 2B$: copies overlap for a general signal occupying the band.

A sinusoid is commonly written as

$$x(t) = A\cos(2\pi f t + \phi).$$

A is amplitude, f is frequency in hertz, t is time in seconds, and ϕ is phase in radians. If the equation uses angular frequency ω, then $\omega = 2\pi f$.

Example 1.11

For $x(t) = 8\cos(200\pi t)$, find the signal frequency, Nyquist rate and Nyquist interval.

SOLUTION

The coefficient of t is 200π rad/s.

Compare with the standard form. $2\pi f = 200\pi$.

Solve for frequency. $f = 200\pi/(2\pi) = 100$ Hz. This single-tone signal has $B = 100$ Hz.

Double the highest frequency. $R_N = 2B = 2(100) = 200$ samples/s.

Take the reciprocal. $T_N = 1/R_N = 1/200 = 0.005$ s $= 5$ ms.

Answer. Frequency: 100 Hz. Nyquist rate: 200 samples/s. Nyquist interval: 5 ms. Choose an actual sampling rate above 200 samples/s.

Key point. Do not use 200π as a frequency in hertz. Divide the angular frequency by 2π first. Amplitude 8 does not change the Nyquist rate.

A sum of sinusoids

When several different sinusoidal components are added, list their frequencies and select the largest one that remains in the signal. Work with the frequency inside each sine or cosine. The number multiplying the whole term is its amplitude.

Example 1.12

Find the Nyquist rate for $x(t) = 3\cos(50\pi t) + 10\sin(300\pi t) - \cos(100\pi t)$.

SOLUTION

Three components with angular frequencies 50π, 300π and 100π rad/s.

Convert each frequency. Divide each coefficient of t by 2π.

Component	Calculation	Frequency
$3\cos(50\pi t)$	$50\pi/(2\pi)$	25 Hz
$10\sin(300\pi t)$	$300\pi/(2\pi)$	150 Hz
$-\cos(100\pi t)$	$100\pi/(2\pi)$	50 Hz

Choose the largest. $B = \max(25, 150, 50) = 150$ Hz.

Apply the sampling rule. $R_N = 2B = 2(150) = 300$ samples/s.

Answer. The Nyquist rate is 300 samples/s. Use an actual rate greater than 300 samples/s for this baseband signal.

Key point. Do not add the component frequencies. The highest frequency sets the baseband limit. The minus sign changes a component's sign, not its frequency.

A product of sinusoids

Multiplication creates new frequencies. Expand a product before finding its highest frequency. The sum-frequency term oscillates faster than either original component, so it determines the required sampling rate in this example.

Example 1.13

Find the Nyquist rate and interval for $x(t) = \cos(4000\pi t)\cos(1000\pi t)$.

SOLUTION

The two original frequencies are 2000 Hz and 500 Hz.

Write the identity. $\cos A \cos B = \frac{1}{2}\cos(A+B) + \frac{1}{2}\cos(A-B)$.

Expand the product.

$$x(t) = \tfrac{1}{2}\cos(5000\pi t) + \tfrac{1}{2}\cos(3000\pi t).$$

Read the new frequencies. $f_1 = 2500$ Hz and $f_2 = 1500$ Hz.

Find the rate. $R_N = 2(2500) = 5000$ samples/s.

Find the interval. $T_N = 1/5000 = 0.0002$ s $= 0.2$ ms.

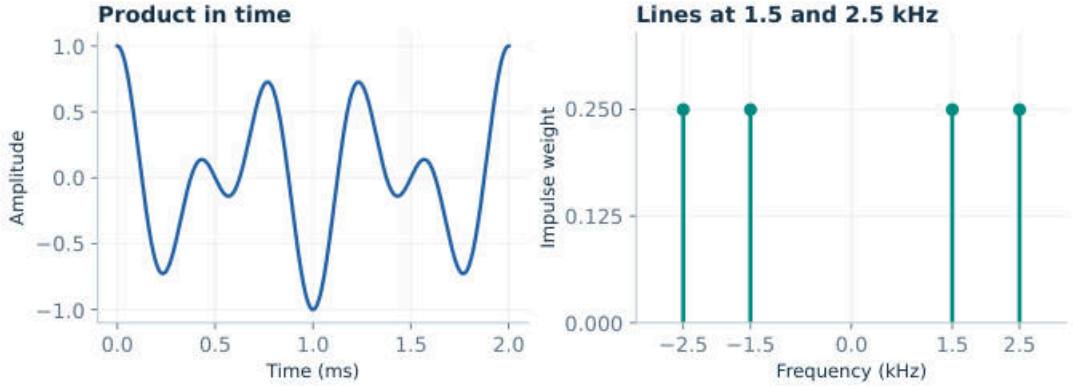


Figure 1.13: Multiplication produces components at the sum and difference frequencies. Each bilateral Fourier impulse has weight 1/4.

Answer. Nyquist rate: 5000 samples/s. Nyquist interval: 0.2 ms.

Key point. For a product, choosing the larger original frequency would give the wrong answer. Expand first.

1.8 Aliasing and the discrete-time sequence

Definition / Aliasing

Aliasing occurs when different continuous-time signals give the same samples. The samples alone cannot distinguish those signals.

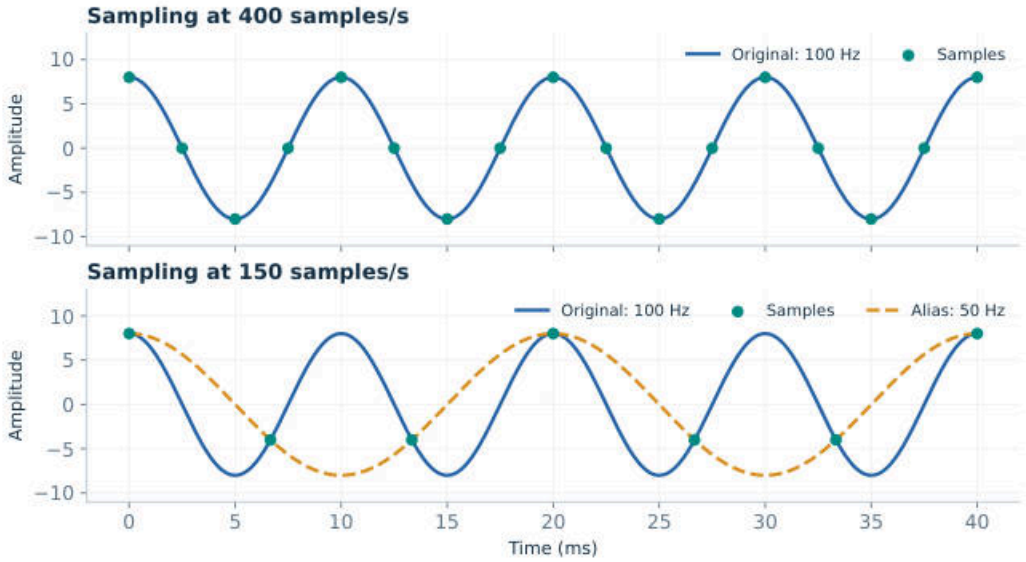


Figure 1.14: At 150 samples/s, the 100 Hz and 50 Hz cosines have the same sample values.

Example 1.14

A 100 Hz cosine is sampled at 150 samples/s. Find the alias frequency between 0 and $f_s/2$.

SOLUTION

$f = 100$ Hz, $f_s = 150$ samples/s. The interval ends at 75 Hz.

Check the sampling rate. $150 < 2(100)$, so this rate is below the Nyquist rate.

Find a frequency in the first interval. $f_a = |f - f_s| = |100 - 150| = 50$ Hz.

Check the interval. $0 \leq 50 \leq 75$ Hz.

Answer. The samples are also consistent with a 50 Hz cosine.

Key point. For a single cosine, choose an integer k so that $f_a = |f - kf_s|$ lies between 0 and $f_s/2$. One subtraction is not enough for every frequency.

To find the sample sequence, replace time t by n/f_s. The sequence is a function of the integer n. For example, $n = 2$ means the measurement at $2/f_s$ seconds. The coefficient of n is an angle per sample, not a frequency in hertz.

Example 1.15

For $x(t) = 8\cos(200\pi t)$, find $x[n]$ at (a) 400 samples/s and (b) 150 samples/s.

SOLUTION

$f = 100$ Hz. Use $x[n] = x(n/f_s)$.

(a) Sampling at 400 samples/s

Substitute the sample time. $x[n] = 8\cos(200\pi n/400)$.

Simplify. $x[n] = 8\cos(\pi n/2)$.

Check the first values. For $n = 0, 1, 2, 3$, the values are $8, 0, -8, 0$. There are four samples per cycle.

(b) Sampling at 150 samples/s

Substitute the sample time. $x[n] = 8\cos(4\pi n/3)$.

Use periodicity for integer n. Subtract $2\pi n$ inside the cosine: $4\pi n/3 - 2\pi n = -2\pi n/3$.

Use cosine symmetry. $\cos(-\theta) = \cos\theta$, so $x[n] = 8\cos(2\pi n/3)$.

Identify the alias. $2\pi f_a/150 = 2\pi/3$, giving $f_a = 50$ Hz.

Answer. At 400 samples/s: $8\cos(\pi n/2)$. At 150 samples/s: $8\cos(2\pi n/3)$, which also represents a sampled 50 Hz cosine.

Key point. The expressions $8\cos(4\pi n/3)$ and $8\cos(2\pi n/3)$ agree at integer n. They are not identical for arbitrary real-valued time.

Example 1.16

A 1700 Hz cosine is sampled at 1000 sample/s. Find the alias in 0 to 500 Hz.
Find: Alias frequency between zero and 500 Hz.

SOLUTION

Try integers k until $|f - kf_s|$ falls in the first Nyquist interval.
$f_a = |f - kf_s|$, test k = 1
k = 1 gives |1700 - 1000| = 700 Hz, outside the interval.
$f_a = |f - kf_s|$, test k = 2

$$k = 2: \quad f_a = |1700 - 2000| = 300\,\text{Hz}.$$

$$0 \le f_a \le f_s/2$$

300 Hz lies between 0 and 500 Hz.

Answer. 300 Hz.
One subtraction of f_s is not always enough. The interval check is essential.

1.9 Anti-alias filtering and practical sampling margin

Definition / Anti-aliasing filter

An analog filter placed before the sampler to reduce frequency components that would fold into the wanted band.

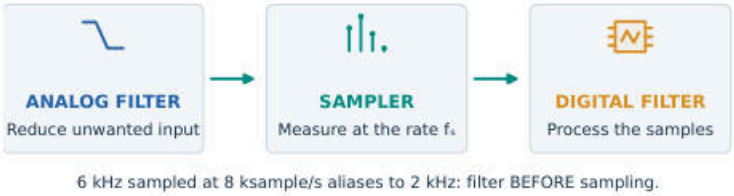


Figure 1.15: The unwanted component must be reduced before sampling.

A digital filter cannot separate two components that have already become indistinguishable through aliasing. The protection therefore starts in the analog path.

Example 1.17

Consider the analog input filtering and sampling arrangement described below. A wanted band ends at 2 kHz. A teaching design requires at least 1 kHz between that edge and $f_s/2$. Find the smallest rate meeting that margin.

Find: Smallest rate satisfying the specified transition-space requirement.

SOLUTION

$$f_s/2 \geq B + M.$$

$$f_{\text{half,required}} = B + M$$

$$B + M = 2 + 1 = 3\,\text{kHz}.$$

$$f_{s,\text{min}} = 2(B + M)$$

$$f_s \geq 2 \times 3 = 6\,\text{ksample/s}.$$

$$f_s/2 - B \geq M$$

At 6 ksample/s, the wanted edge is 2 kHz and half-rate edge is 3 kHz.

Answer. Minimum for the stated margin: 6 ksample/s.
This rate satisfies the given geometric margin. A practical filter still needs passband and stopband specifications.

1.10 Spectral replicas and the sampling-condition derivation

A spectrum describes which frequencies are present. Let $X(f)$ be the Fourier transform of $x(t)$, using frequency f in hertz.

Mark every sampling instant. An ideal impulse train is

$$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT_s).$$

The impulse $\delta(t - nT_s)$ marks the instant nT_s.

Select the signal values. Multiply by the impulse train:

$$x_s(t) = x(t)p(t) = \sum_{n=-\infty}^{\infty} x(nT_s)\delta(t - nT_s).$$

The weight of each impulse is the corresponding sample value.

Use the impulse-train transform. Its transform is another impulse train:

$$P(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} \delta(f - kf_s).$$

The integer k identifies a spectral copy. For $k = 0$, the copy stays at zero. For $k = 1$, its center moves to f_s. For $k = -1$, its center moves to $-f_s$.

Apply multiplication–convolution duality. Multiplication in time becomes convolution in frequency:

$$X_s(f) = X(f) * P(f) = \frac{1}{T_s} \sum_{k=-\infty}^{\infty} X(f - kf_s).$$

Here $*$ denotes convolution. Each term is a shifted copy of $X(f)$.

Require a gap between copies. The original ends at B. The next copy starts at $f_s - B$. Separation requires

$$B < f_s - B \quad \Longrightarrow \quad f_s > 2B.$$

> **Key point.** The algebra leads to the same picture shown earlier: copies are spaced by f_s, and each baseband copy extends from $-B$ to B.

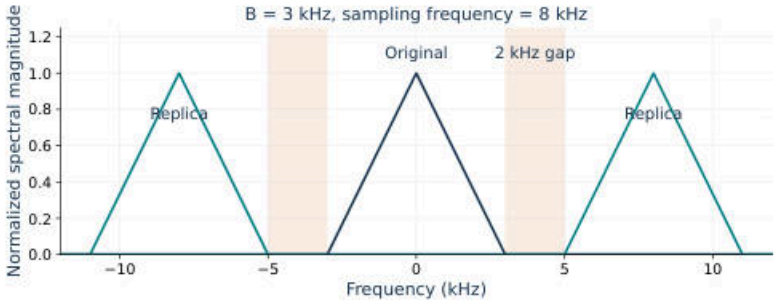


Figure 1.16: For a 3 kHz baseband limit and 8 ksample/s sampling, the first positive replica begins at 5 kHz, leaving a 2 kHz separation gap.

1.11 Ideal reconstruction, held output and squared sinc

The reconstruction filter keeps the original spectral copy and rejects the others. Its gain also removes the scale factor $1/T_s$ introduced by sampling.

Choose an ideal low-pass filter. For $f_s > 2B$, choose

$$H(f) = \begin{cases} T_s, & |f| < f_s/2, \\ 0, & |f| > f_s/2. \end{cases}$$

Inside the original band, $T_s X_s(f) = X(f)$.

Find its impulse response. The inverse Fourier transform gives

$$h(t) = T_s \int_{-f_s/2}^{f_s/2} e^{j2\pi ft}\, df = T_s \frac{\sin(\pi f_s t)}{\pi t} = \operatorname{sinc}(t/T_s).$$

Here $j^2 = -1$, $f_s T_s = 1$, and $\operatorname{sinc}(u) = \sin(\pi u)/(\pi u)$ with $\operatorname{sinc}(0) = 1$.

Add the response to every sample impulse. Convolving $x_s(t)$ with $h(t)$ gives

$$\boxed{x(t) = \sum_{n=-\infty}^{\infty} x[n] \operatorname{sinc}\left(\frac{t - nT_s}{T_s}\right).}$$

Each sample sets the height of one shifted sinc pulse.

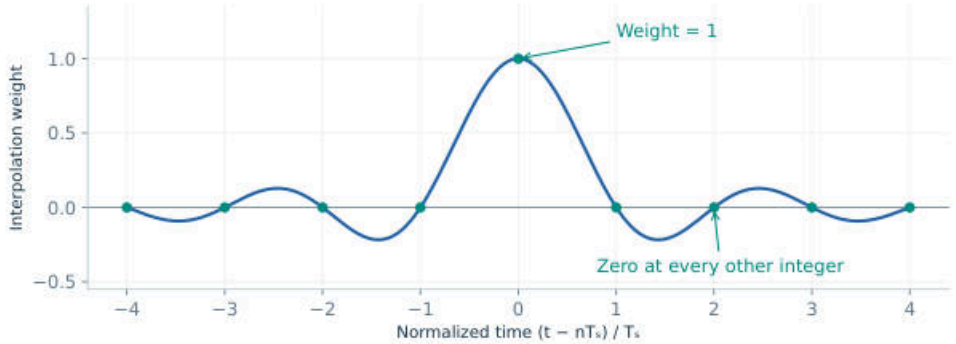


Figure 1.17: A sinc pulse is 1 at its own sample instant and 0 at the other integer sample positions.

Key point. At $t = mT_s$, only the term $n = m$ remains, so the formula returns $x[m]$. The ideal formula uses infinitely many samples. Practical filters approximate it.

Example 1.18

Consider reconstruction from uniformly spaced samples using the specific reconstruction model stated below. Only x[0] = 1 V and x[1] = 0.5 V are nonzero; all other samples are zero. Find the two-term interpolation at t = 0 and t = T_s.

Find: Interpolated values at the two specified instants.

SOLUTION

At an integer sample instant, all sinc terms except the matching one are zero.

$$y(0) = x[0]\,\mathrm{sinc}(0) + x[1]\,\mathrm{sinc}(-1)$$

$$y(0) = \mathrm{sinc}(0) + 0.5\,\mathrm{sinc}(-1) = 1 + 0 = 1\,\mathrm{V}.$$

$$y(T_s) = x[0]\,\mathrm{sinc}(1) + x[1]\,\mathrm{sinc}(0)$$

$$y(T_s) = \mathrm{sinc}(1) + 0.5\,\mathrm{sinc}(0) = 0 + 0.5 = 0.5\,\mathrm{V}.$$

Answer. $y(0) = 1\,\mathrm{V}; y(T_s) = 0.5\,\mathrm{V}$.
This is a fully specified two-nonzero-sample example, not a generic truncation claimed to be exact for every signal.

Example 1.19

Consider reconstruction from uniformly spaced samples using the specific reconstruction model stated below. A zero-order hold receives 1 V at t = 0 and -0.5 V at t = 1 ms. Find its output at 0.6 ms and 1.2 ms, assuming the next sample occurs at 2 ms.

Find: Held output at the two specified times.

SOLUTION

A zero-order hold keeps each sample value until the next sample arrives.

$$y_{\mathrm{hold}}(t) = x[n], nT_s \le t < (n+1)T_s$$

0.6 ms is in the first interval, so the held output is 1 V.
1.2 ms is in the second interval, so the held output is -0.5 V.

Answer. 1 V at 0.6 ms; -0.5 V at 1.2 ms.
A held staircase and an ideal sinc-interpolated curve are different reconstruction descriptions.

Squared sinc and spectral support

Example 1.20

For $x(t) = \text{sinc}^2(200t)$, find the baseband bandwidth, Nyquist rate and Nyquist interval. Use normalized sinc.

SOLUTION

$\text{sinc}(u) = \sin(\pi u)/(\pi u)$.

Start with the unsquared sinc. The transform of $\text{sinc}(200t)$ is a rectangle extending from -100 to $+100$ Hz. Its scale factor is $1/200$.

Apply the square. Squaring in time convolves the rectangle with itself in frequency.

Add the frequency limits. The new support extends from $-100 - 100 = -200$ Hz to $100 + 100 = 200$ Hz. Its shape is triangular.

Read the baseband bandwidth. $B = 200$ Hz, the positive upper edge.

Calculate rate and interval. $R_N = 2B = 400$ samples/s and $T_N = 1/400 = 2.5$ ms.

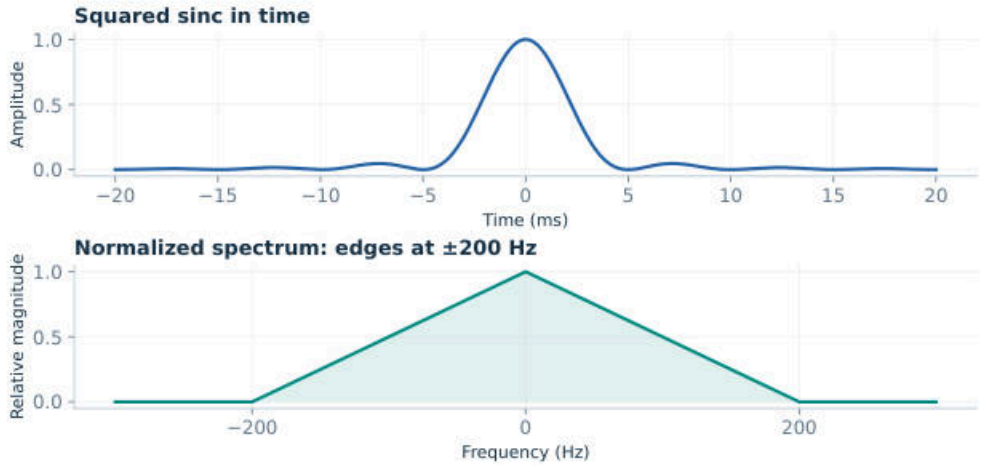


Figure 1.18: The squared-sinc waveform has a triangular spectrum with edges at ± 200 Hz.

Answer. Bandwidth: 200 Hz. Nyquist rate: 400 samples/s. Nyquist interval: 2.5 ms.

Key point. The total spectral width is 400 Hz, but the baseband bandwidth is 200 Hz. Squaring the waveform does not mean simply doubling its time argument.

Build reconstruction from individual samples

Use Example 1.18: $x[0] = 1$ V and $x[1] = 0.5$ V, with all other samples zero. Let $u = t/T_s$ and $\mathrm{sinc}(u) = \sin(\pi u)/(\pi u)$, with value 1 at zero.

$$y(t) = 1\,\mathrm{V}\,\mathrm{sinc}(u) + 0.5\,\mathrm{V}\,\mathrm{sinc}(u-1).$$

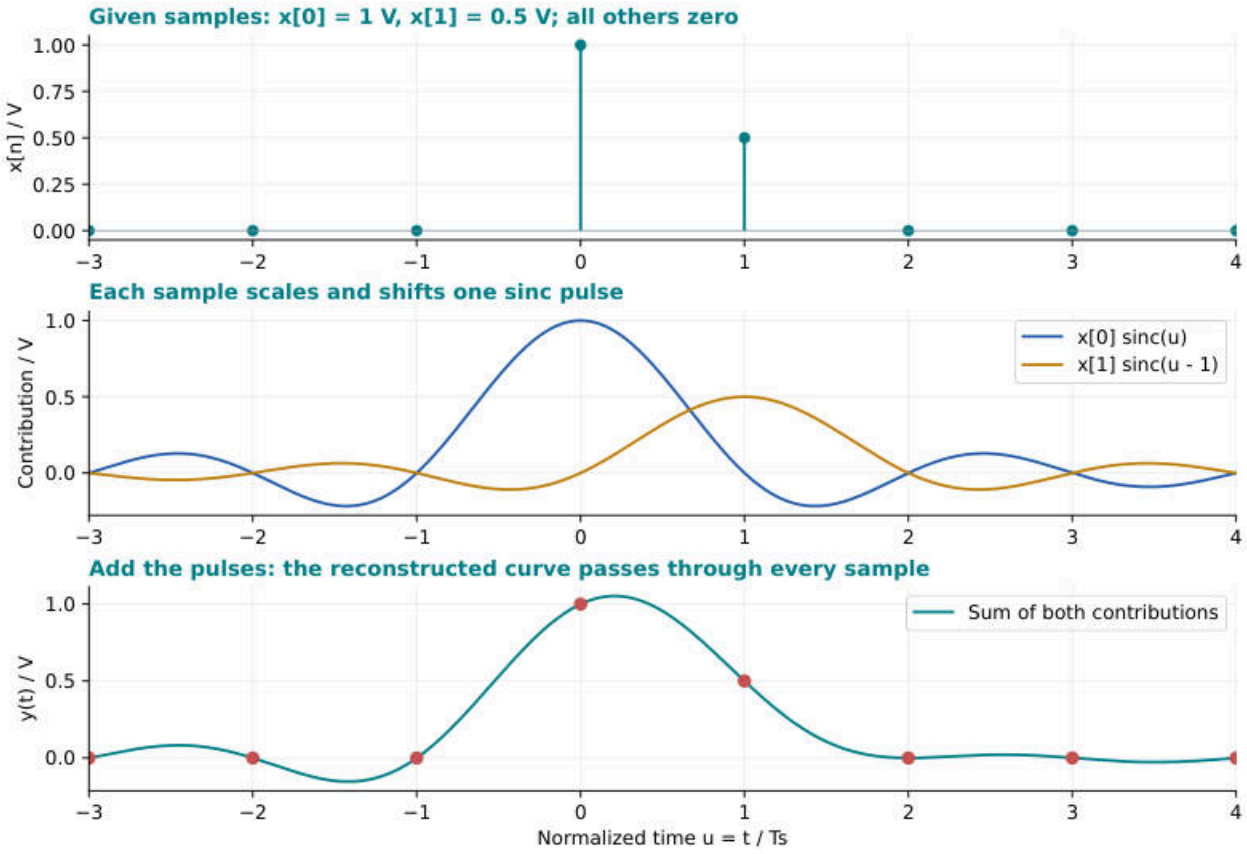


Figure 1.19: Scale and shift one sinc pulse for each nonzero sample, then add the contributions.

Read the sum. At $u = 0$, blue contributes 1 V and gold contributes zero. At $u = 1$, blue contributes zero and gold contributes 0.5 V. At every other integer both vanish. This is exact for this specified sequence, not for an arbitrary signal truncated to two samples.

1.12 Bandpass sampling windows and spectral inversion

Definition / Bandpass signal

A bandpass signal occupies a band away from zero frequency. Write its positive-frequency edges as f_L and f_H. Its occupied bandwidth is $B = f_H - f_L$.

Sampling above $2f_H$ is a straightforward safe choice. A lower rate may also work if the shifted positive and negative bands do not overlap.

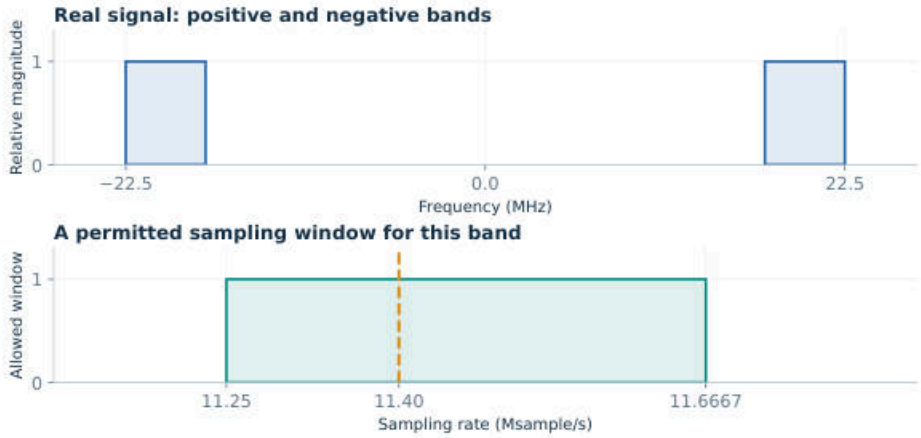


Figure 1.20: An occupied band away from zero can fit into a suitable sampling window.

How to test a lower sampling rate

For an ideal band-limited real signal, a candidate window is

$$\frac{2f_H}{m} \le f_s \le \frac{2f_L}{m-1}, \qquad 2 \le m \le \left\lfloor \frac{f_H}{B} \right\rfloor.$$

- m is an integer that selects a possible sampling window.
- $\lfloor \cdot \rfloor$ means round down to an integer.
- Calculate both ends of the window before choosing f_s.
- Use an interior value in practice, allowing filter and frequency margins.

Key point. Do not use $f_s = 2B$ automatically for a bandpass signal. The band location matters as well as its width. If no $m \ge 2$ is available, use the ordinary rate above $2f_H$.

Example 1.21

A real signal occupies 17.5 to 22.5 MHz on its positive-frequency side. Find a permitted lower sampling-rate window and select a rate inside it.

SOLUTION

$f_L = 17.5$ MHz and $f_H = 22.5$ MHz.

Calculate the occupied bandwidth. $B = 22.5 - 17.5 = 5$ MHz.

Find the largest permitted integer. $\lfloor f_H/B \rfloor = \lfloor 22.5/5 \rfloor = 4$. Choose $m = 4$ to test the lowest candidate window.

Calculate the lower limit. $2f_H/m = 2(22.5)/4 = 11.25$ MHz.

Calculate the upper limit. $2f_L/(m-1) = 2(17.5)/3 = 11.6667$ MHz.

Choose an interior rate. Use $f_s = 11.4$ Msample/s, which lies between the two limits.

Answer. A candidate window is $11.25 \le f_s \le 11.6667$ Msample/s. The choice 11.4 Msample/s is inside it.

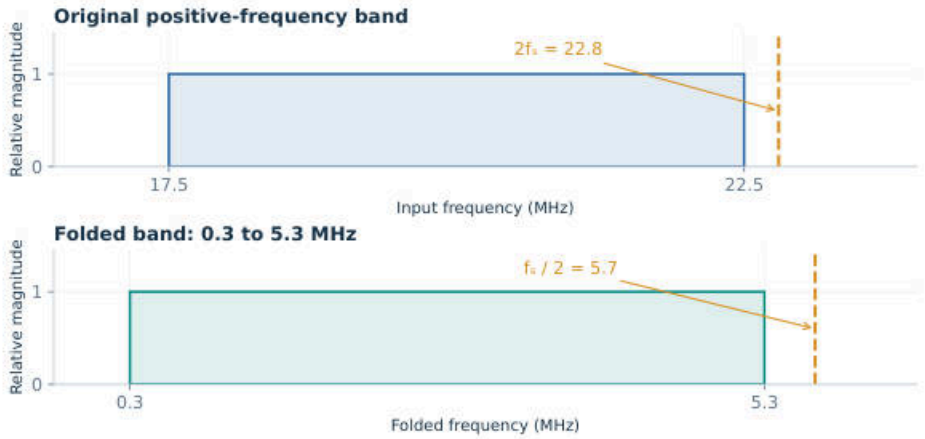


Figure 1.21: At 11.4 Msample/s, the positive RF band folds into 0.3–5.3 MHz, inside the interval 0–5.7 MHz.

Check the folded limits

$2f_s - f_H = 22.8 - 22.5 = 0.3$ MHz and $2f_s - f_L = 22.8 - 17.5 = 5.3$ MHz. Both lie below $f_s/2 = 5.7$ MHz. The order reverses, so the folded band is spectrally inverted. This means that the original upper edge becomes the lower folded edge. A receiver must account for this reversal when interpreting the spectrum.